YEAR 1 · TERM 1 · MODULE 3 · WEEK 1

Lab Workbook

Database management systems

Level 1 · 25 min | Level 2 · 30 min | Level 2 · 25 min | Level 3 · 40 min

SE Study Lab · Study material

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Run these queries in the subject's SQL Lab: a real database in your browser, with a button to reset the data.

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Do this lab in the subject's Design Studio, then export the diagram to draw.io.

1 Level 1 · 25 min
First queries in the SQL Lab

Goal: Run SELECT, WHERE, and ORDER BY on the campus store database.

1. List every product

```
SELECT * FROM product;
```

Expected result: Five products with their id, name, price, and stock.

Reflect: What changed after this step, and why?

2. Filter by price

```
SELECT name, price FROM product WHERE price > 20000;
```

Expected result: Only products that cost more than 20,000.

Reflect: What changed after this step, and why?

3. Sort the result

```
SELECT name, price FROM product ORDER BY price DESC;
```

Expected result: All products, from the most to the least expensive.

Reflect: What changed after this step, and why?

2 Level 2 · 30 min
Join customers and orders

Goal: Use JOIN to answer "what did Ana buy?"

1. Customers with their orders

```
SELECT c.name, o.order_id
FROM customer AS c
JOIN orders AS o ON c.customer_id = o.customer_id;
```

Expected result: Ana appears twice because she has two orders.

Reflect: What changed after this step, and why?

2. What did Ana buy?

```
SELECT p.name, oi.quantity
FROM customer AS c
JOIN orders AS o ON o.customer_id = c.customer_id
JOIN order_item AS oi ON oi.order_id = o.order_id
JOIN product AS p ON p.product_id = oi.product_id
WHERE c.name = 'Ana';
```

Expected result: The products in Ana's orders, with quantities.

Reflect: What changed after this step, and why?

Level 2 · 25 min

3 See integrity and transactions at work

Goal: Trigger constraint errors on purpose, and undo a change with ROLLBACK.

1. Break the foreign key

```
INSERT INTO orders (order_id, customer_id, order_date) VALUES (200, 99, '2026-03-01');
```

Expected result: An error: customer 99 does not exist (FOREIGN KEY constraint failed).

Reflect: What changed after this step, and why?

2. Break the CHECK rule

```
UPDATE product SET price = -5 WHERE product_id = 10;
```

Expected result: An error: the CHECK constraint forbids negative prices.

Reflect: What changed after this step, and why?

3. Undo with ROLLBACK

```
BEGIN;
DELETE FROM order_item;
SELECT COUNT(*) FROM order_item;
ROLLBACK;
SELECT COUNT(*) FROM order_item;
```

Expected result: The count drops to 0 inside the transaction, then returns to its original value after ROLLBACK.

Reflect: What changed after this step, and why?

Level 3 · 40 min

4 Design a library in the Design Studio

Goal: Build the library model, check it, then open it in draw.io to polish the diagram.

1. Choose the entities
Expected result: In the Design Studio, pick the Library scenario and add Member, Book, and Loan. Leave out the distractors.

Reflect: What changed after this step, and why?

2. Set keys and relationships
Expected result: Mark one primary key per entity, and connect Member 1:N Loan and Book 1:N Loan.

Reflect: What changed after this step, and why?

3. Check and export
Expected result: Press Check design until every item is green, then Export to draw.io and open the file at app.diagrams.net.

Reflect: What changed after this step, and why?

Troubleshooting

Symptom	Likely cause	Fix
FOREIGN KEY constraint failed	The referenced row does not exist	Insert the parent row first (for example, the customer)
CHECK constraint failed	A value breaks a rule, such as a negative price	Fix the value or the rule
Every row changed	UPDATE or DELETE without WHERE	Test the condition with SELECT first; work inside a transaction